

Route-Complete Evidence Package

Dual-Route Admissible Execution Evidence — TA-14 Framework

Package ID	OMNIX-EP-76A900F4A33640BC
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OMNIX Version	2.6.0
Scenario	QuantumBank AI Trading Desk
Agent	OMNIX-AGENT-QUANT-001
Action	Authorize \$2,000,000 capital reallocation under RISK-MANDATE-2026-Q2
Domain	capital_markets
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
Standard	NIST FIPS 204
Security Level	NIST Level 3

TA-14 Chain Map

Every link in the admissible execution chain is mapped to a PQC-signed OMNIX artifact.

TA-14 Step	OMNIX Artifact
Reality	1_source_state — source_state_hash
Record	2_record — DelegationReceipt (ATFDR-*) — PQC-signed
Continuity	3_continuity — RCR continuity record — CES score — PQC-signed
Admissibility	4_admissibility — TemporalAdmissibilityRecord (ATFTAR-*) — PQC-signed
Binding	5_binding — binding_record — authority basis accepted — PQC-signed
Commit	6_commit — locked scope — execution_reachable flag — PQC-signed
Execution	7_execution — BAR (behavioral output) + CTCHC (turn chain) — PQC-signed
Outcome	8_outcome — PoGC (proof certificate) + outcome_receipt — PQC-signed

Route A — Hard Refusal

TA-14 Verdict: REFUSAL PROVEN — execution structurally blocked at admissibility gate

Step	Artifact Type	Status / Key Fields
1	1_source_state	—
2	2_record	ACTIVE
3	3_continuity	—
4	4_admissibility	ADMITTED ATFTAR-64495F309A1344C3

5	5_binding	REFUSED ATFTAR-64495F309A1344C3
6	6_commit	BLOCKED
7	7_execution	—
8	8_outcome	—

Route B — Full Admission

TA-14 Verdict: **ADMISSION PROVEN** — complete authority chain from source state to outcome

Step	Artifact Type	Status / Key Fields
1	1_source_state	—
2	2_record	ACTIVE
3	3_continuity	— ATFRCR-7583ED1BD5BA4E73
4	4_admissibility	ADMITTED ATFTAR-B22C96B0309D4D60
5	5_binding	ACCEPTED ATFTAR-B22C96B0309D4D60
6	6_commit	LOCKED
7	7_execution	—
8	8_outcome	—

Invariants Demonstrated

Invariant ID	Formal Statement
ATF-INV-001	Authority never expands through delegation (MAR enforced at DR issuance)
ATF-INV-002	Every DR is PQC-signed by the delegator
ATF-INV-005	Receipts are immutable once issued (content_hash)
TAR-INV-001	TAR issued synchronously at admission time
TAR-INV-002	execution_ns captured by time.time_ns() at this call
TAR-INV-003	Expired or exhausted DRs produce admission_status=REJECTED
RGC-INV-001	Every RCR anchored to a valid TAR
RGC-INV-002	CES computed from real-time values only
BEV-INV-001	Every governed turn produces a BAR before output is delivered
BEV-INV-002	BAR content_hash covers output_hash + governing_receipt_id + turn_index
BEV-INV-004	BAR PQC signature verifiable offline without calling OMNIX
BEV-INV-010	CTCHC initialized before first BAR
BEV-INV-011	Each CTCHC link = H(prev turn_hash governing_receipt_id)
BEV-INV-013	Seal hash covers complete chain (first→last link)

Offline Verification

Command:

```
python scripts/verify_evidence_package.py omnix_evidence_package_20260527_191542.json
```

What is verified:

- DR content_hash integrity (SHA-256 of all fields)
- DR PQC signature (Dilithium-3 over content_hash)
- TAR content_hash integrity
- TAR PQC signature
- RCR hash integrity
- RCR PQC signature
- Binding record hash + PQC signature
- Commit record hash + PQC signature
- BAR content_hash (SHA3-256 covering output_hash + receipt_id + turn_index)
- BAR PQC signature
- CTCHC link-by-link hash chain integrity
- CTCHC seal_hash covers all links in order
- CTCHC seal PQC signature
- PoGC content_hash integrity
- PoGC PQC signature
- Refusal receipt hash + PQC signature (Route A)
- Outcome receipt hash + PQC signature (Route B)

Verifier scope limits:

- This verifier confirms cryptographic integrity and hash chain consistency.
- It does NOT verify governance policy parameters (mandate values, risk ceilings) — those require docs/standards/.
- It does NOT verify external market data referenced in source_state.
- It DOES confirm that the public key in this package produces valid signatures for all artifacts.